Direct training reaches good QFT operators

Part 1 (`1_structure_inclusion/`) showed the trained blocked optimum lives **inside** the QFT family, reachable by warm-start. This part asks the complementary question: can good QFT operators be found by training **directly** — without the blocked solution to start from? Two lines of evidence say yes. Both train on DIV2K-8q ($m = n = 8$) under the headline preset (1008 steps, seed 42), so they compare directly against the headline `qft` cell (31.29 dB at $\rho = 0.20$) and the `blocked_8` reference optimum (32.26 dB).

Identity init and L1 regularisation (`identity_l1/`)

The analytic-QFT prior is not required. Initialising every gate at its manifold identity ($T_{t=0}(x) = x$) — same gate topology as `qft`, no analytic phase values — and training under the identical budget reaches 31.66 dB at $\rho = 0.20$, **0.37 dB above** the analytic-init `qft`. What carries the result is the gate topology (which qubits the Hadamards and controlled-phase gates act on), shared by both inits; the analytic phases are incidental. (Detail: `identity_l1/writeup.pdf`.)

L1-to-identity regularisation finds dataset-adaptive sparse circuits. Adding a sparsity term that pulls gates toward identity, $\mathcal{L} = \text{MSE-topK}(\theta) + \lambda \cdot R(\theta)$, swept over λ, turns training into a search for a **sparse** QFT operator. On DIV2K-8q the regularised optimum matches `blocked_8` with a structurally **different** sparse circuit; on sparse QuickDraw sketches it collapses to the **identity** basis, beating every block transform by tens of dB. A block-masked L2 companion behaves analogously. So direct training, suitably regularised, **discovers** compression-appropriate structure rather than being handed it. (Detail: `identity_l1/writeup_reg_sweep.pdf`; the L1-init-anchor transfer test across other topologies lives in the same cell tree.)

Progressive gate-unfreezing (`unfreeze/`)

Rather than train all 72 gates jointly, thaw **one gate per stage** and train each to a plateau before unfreezing the next. This curriculum reaches **deeper** into the family than joint training:

unfreeze order	**identity init**	**random init**	
block-growth (`bg`)	**31.98**	31.66	PSNR@$\rho = 0.20$ (dB)
left→right (`lr`)	31.84	31.66	
right→left (`rl`)	30.81	30.81	

The best curriculum (block-growth from identity, 31.98 dB) clears the headline joint-trained `qft` by 0.69 dB and approaches the blocked optimum. The **order** matters — growing a square block (`bg`) or following the QFT construction order (`lr`) both beat the reversed order (`rl`) by ~1 dB, and `rl` is insensitive to initialisation while `bg`/`lr` benefit from the identity start. (Detail + per-stage staircase dynamics: `unfreeze/writeup.pdf`.)

Takeaway

Good QFT operators are **trainable directly**, given the right protocol. The structural prior that matters is the gate topology, not the analytic phase values: from a featureless identity init, plain Adam (31.66 dB), L1-regularised search (dataset-adaptive sparse circuits), and gate-by-gate unfreezing (31.98 dB) all reach operators at or above the headline `qft`, climbing toward the blocked optimum that part 1 placed inside the family. Part 3 (`3_training_topk/`) then shows the training objective itself — the top-k truncation rate — is a further lever.